



Workshop::Cluster Creation

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Common Issues in Cluster Creation #1

- Cluster creation may be stuck “in progress” for hours, which suggests there are underlying issues
- Not enough resources: some VMs may have been created for practice and not have been removed before creating the cluster (a project has only 11 vCPUs, and the cluster takes 8 of them).

To diagnose use of resources, use the MRC dashboard or `openstack server list --fit-width`

- Floating IP address taken: every project has only one floating IP address (a public IP address), and the cluster needs one.
To check how many addresses have been used, go to the MRC dashboard or: `openstack floating ip list --fit-width`

Common Issues in Cluster Creation #2

- The cluster was created, but when the kubeconfig file has to be generated with `openstack coe cluster config comp90024`, the command returns “Policy doesn't allow certificate:get to be performed (HTTP 403)”
- This means that the cluster has not been created by you: only the team member who created the cluster can generate the kubeconfig file
- Only one cluster can be created per project given the cluster template used in this course.
An attempt to create a cluster when there is another one created or in the process of creation/deletion would fail

Access to the Swift Object Store Service

- The OpenStack Swift service is an AWS S3-compatible object store that allows to write and read data using HTTP
- This service can be used, for instance, to store Elasticsearch backups, or store data that can be then be loaded, by pods in the Kubernetes cluster, onto Elasticsearch
- However, trying to add a file to a Swift container, as in `openstack object create <container> <file>` may return error HTTP 413: Request Entity Too Large
- This is due to MRC policies. Just send an email to the teaching staff and we will add some storage quota to your project.
By default your project has 0 storage quota for Swift